GRADE 7 ACCELERATED · UNIT 4 · LESSON 1

Adding and Subtracting in Scientific Notation

Question Bank 4.01 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 6, add or subtract. Write each answer in scientific notation.

1. $(2.4\times10^{5})+(3.1\times10^{5})$

5.5×10^{5}

2. $(8.3\times10^{-4})-(2.1\times10^{-4})$

6.2×10^{-4}

3. $(6.7\times10^{9})+(5.8\times10^{9})$

$12.5\times10^{9}=1.25\times10^{10}$

The sum of the coefficients is over 10, so the answer needs one more step.

4. $(9.1\times10^{4})-(8.6\times10^{4})$

$0.5\times10^{4}=5\times10^{3}$

A coefficient below 1 is unfinished too: point one place right, exponent down one.

5. $(7.2\times10^{6})+(4.3\times10^{5})$

7.63×10^{6}

$4.3\times10^{5}=0.43\times10^{6}$ *, and* $7.2+0.43=7.63$.

6. $(5.5\times10^{-3})-(8.0\times10^{-5})$

5.42×10^{-3}

$8.0\times10^{-5}=0.08\times10^{-3}$. *The exponents are two apart, the widest gap allowed.*

Real quantities.

7. Neptune's diameter is about 4.95×10^4 kilometers. Saturn's is about 1.21×10^5 kilometers.

a. What is the sum of the two diameters?

1.705×10^5 ***km***

$1.21 \times 10^5 = 12.1 \times 10^4$ *, so the sum is* 17.05×10^4 .

b. Jupiter's diameter is about 1.43×10^5 kilometers. Is your answer to part (a) greater than Jupiter's diameter?

Yes — $1.705 \times 10^5 > 1.43 \times 10^5$ **.**

Same power of ten, so the coefficients decide.

8. A city's population is about 4.62×10^5 . The surrounding county's population is about 1.38×10^6 .

a. What is the total population of the city and the county?

1.842×10^6 ***people***

$4.62 \times 10^5 = 0.462 \times 10^6$.

b. How many more people live in the county than in the city?

9.18×10^5 ***people***

$1.38 \times 10^6 = 13.8 \times 10^5$ *, and* $13.8 - 4.62 = 9.18$.

Working backwards.

9. Find the missing number in each equation.

a. $(3.6 \times 10^7) + (n \times 10^7) = 1.01 \times 10^8$

n = 6.5

$1.01 \times 10^8 = 10.1 \times 10^7$.

b. $(8.4 \times 10^{-6}) - (n \times 10^{-6}) = 2.9 \times 10^{-6}$

n = 5.5

c. $(2.5 \times 10^4) + (7.5 \times 10^4) =$ _____

1×10^5

The coefficients sum to exactly 10, so the answer is 10×10^4 *— finished, that is* 1×10^5 .

Reading someone else's work.

10. Priya writes $(2.6\times10^4)+(5.1\times10^6)=7.7\times10^{10}$. Name the two mistakes she made, and give the correct sum.

She added the coefficients without first writing both numbers over the same power of ten, and she added the exponents. The correct sum is 5.126×10^6 ***.***

$2.6\times10^4=0.026\times10^6$ *, and* $5.1+0.026=5.126$.

11. Which is the value of $(4.7\times10^{-5})+(8.6\times10^{-5})$, written in scientific notation?

- ○ A. 13.3×10^{-10}
- ● **B.** 1.33×10^{-4}
- ○ C. 1.33×10^{-10}
- ○ D. 13.3×10^{-5}

B

12. Select ALL of the expressions whose value is 1.2×10^5 .

- ☒ **A.** $(1.25\times10^5)-(5\times10^3)$
- ☐ B. $(1.3\times10^5)-(1\times10^5)$
- ☒ **C.** $(1.3\times10^5)-(1\times10^4)$
- ☐ D. $(8\times10^4)+(4\times10^5)$
- ☒ **E.** $(8\times10^4)+(4\times10^4)$
- ☒ **F.** $(1.1\times10^5)+(1\times10^4)$

A, C, E and F

E gives 12×10^4 *, which is* 1.2×10^5 *once it is finished.*